LOCALISATION PRESERVES EXACTNESS OF SEQUENCES OF MODULES

THE STACKS PROJECT AUTHORS

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Editorial note. The counted statement contains a slogan environment, which the chapter's own preamble defines as a comment environment; the preamble spans of this card start at line 8 and therefore cover it.

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Allowed standard prerequisites (editorial).

- Localisation as a functor. Used at the first line of the counted proof (line 1563). Immediately gives that the localised sequence is a complex, since a functor preserves composition and zero maps.
- Exactness in the middle tested on fractions. Used in the second half of the counted proof (lines 1564-1568). Lets an element of S^{-1}M mapping to zero be represented so that its numerator can be pushed into the kernel of v.

Proposition 0.1. *Let $L \xrightarrow{u} M \xrightarrow{v} N$ be an exact sequence of R-modules. Then $S^{-1}L \to S^{-1}M \to S^{-1}N$ is also exact.*

Proof. First it is clear that $S^{-1}L \to S^{-1}M \to S^{-1}N$ is a complex since localization is a functor. Next suppose that x/s maps to zero in $S^{-1}N$ for some $x/s \in S^{-1}M$. Then by definition there is a $t \in S$ such that $v(xt) = v(x)t = 0$ in N, which means $xt \in \mathrm{Ker}(v)$. By the exactness of $L \to M \to N$ we have $xt = u(y)$ for some y in L. Then x/s is the image of y/st. This proves the exactness. □

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